Skin Lesion Classification

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Technical Updates

WHAT WE TRIED

- Other models (ResNet50, EfficientNetB0, CatBoost)
- Data augmentations geared towards skin cancer classification
- Engineered features like border irregularity and diameter

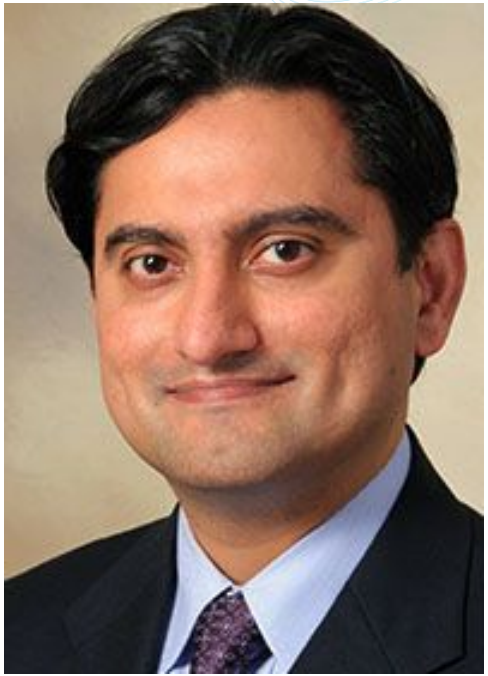
WHAT WE FOUND

- ResNet50 performed slightly better than MobileNetV2
- Rare classes with limited training data are the bottleneck
- XGBoost outperformed deep learning (79% vs 52%)
- Engineered features like border irregularity and diameter estimate were consistently the most influentialfeatures

Persona Overview

President of American Academy of Dermatology Association (AADA)

- Organization has 21,000 members worldwide
- AADA is dedicated to advancing the field of dermatology and improving patient care through improvements in diagnosis, surgery, treatment, and technology
- Murad Alam has a background and interest in skin cancer treatment
- As president of the AADA, he wants to adopt new technologies



Murad Alam, MD, MSCI, MBA, FAAD

Vice Chair, Department of Dermatology
Chief of Cutaneous and Aesthetic Surgery in the Department of Dermatology
Professor, Dermatology (Cutaneous and Aesthetic Surgery), Medical Social Sciences (Outcome and Measurement Science), Otolaryngology - Head and Neck Surgery, Surgery (Organ Transplantation)

Problem & Significance

The Problem:

1. Manual diagnostics can be time-consuming and dependent on specialist skill
2. Current deep learning models have limited explainability

Why it matters:

1) Early detection and treatment is essential for skin cancer treatement
2) AI tools could support faster screening

Stake Holders & Impact

Patients

1) Early detection improve patient outcomes
2) Fast Screening
3) Improved Access to Care

Dermotologist

1. Faster triage
2. Workflow support
3. Second - Opinion Assistance
4. Clinical oversight

Healthcare Facilities

1. Efficiency
2. Lower costs
3. Scalable screening support

What Data Can & Cannot Tell Us

Detect Meaningful Patterns

1. The most important lesion features
2. Performs well on common classes

Rare Classes Need More Data

1. Class imbalance is the main issue
2. Similar lesions confused
3. More diverse data

Improves Clinical Decision Support

1. Serve as a screening tool
2. Catch cases that dermatologists don't

Cannot Replace Clinical Judgment

1. Patient History
2. Physical Exams
3. Biopsy Results

Key Insights

Machine learning out performed Deep learning

Deep learning isn't always the best choice in every case. It fails when using more realistic data

Data quality mattered more than model complexity

More balanced, higher-quality data would likely improve results faster than switching models.

Explainability increases trust

XAI helping doctors review suspicious cases faster.

Recommendation and Impact

Deploy as Descision Support

Deploy the best performing model as a first-pass screening tool to flag high-risk lesions for priority clinical review within dermatology workflows.

Improve Clinical Efficiency

Automated lesion screening allows dermatologists to focus on the highest risk cases, enabling earlier intervention for malignant lesions.

Build Trust Through Explanability

SHAP and Grad-CAM give clinicians full transparency into how each prediction was made, maintaining physician authority in the diagnostic process.

Long Term Strategic Value

The pipeline is modular and lightweight. As more data is collected, all models can be continuously retrained and improved without expensive infrastructure.

Confidence & Uncertainty

Validations

Both models successfully learned meaningful visual patterns from dermoscopic images. SHAP and Grad-CAM confirmed that clinically relevant features including lesion size, border irregularity, and color contrast are consistently driving predictions across both approaches, aligning with established dermatological assessment criteria.

Uncertainties

Performance on rare lesion types remains unreliable due to limited training data. Models may not generalize to images captured with different equipment or patient populations outside the training distribution.

Risk & Consequences

Over Predictions

- Models may flag benign lesions as malignant, leading to unnecessary follow-up appointments and patient anxiety.
- The consequence is manageable as a dermatologist reviews and rules out malignancy.
- The cost is time, not patient safety.

Under Predictions

- The more serious risk is a malignant lesion being misclassified as benign, particularly for underrepresented classes both models struggle with.
- Melanoma misclassification carries the highest clinical risk and is why the system must always support, never replace, clinical judgment.

Mitigations

- All predictions require clinician review before any action is taken.
- A confidence threshold should be set so low-confidence predictions are always escalated to a specialist.
- Models must be retrained as new data becomes available with clear disclaimers displayed to clinicians and patients.

Thanks!

Now for Deployment!